

Bovnar — EBNF Grammar

Bovnar (BVNR) v1.1 documentation · EBNF Grammar · 2026-06-16

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(* ===== *)
(* BOVNAR - EBNF derived from parser/lexer/validator implementation *)
(* Version 1.1 (draft) Notation: ISO/IEC 14977:1996 *)
(* { } = zero or more [ ] = optional *)
(* ( ) = grouping | = alternation *)
(* , = concatenation ; = rule terminator *)
(* "..." = literal terminal (*...*) = comment *)
(* Updated: 2026-06-15 - spec 1.1 (additive over the frozen 1.0): *)
(* version directive (!bovnr M.N), datetime type family, *)
(* \xHH / \u{...} string escapes, reference array indexing. *)
(* Earlier (2026-05-29): null/true/false/on/off keywords, bool *)
(* type family, special numbers respelled nan / inf / ninf *)
(* ===== *)

(* ----- *)
(* 1. Top-level stream *)
(* ----- *)

(* Valid end-of-stream states (lexer states where EOF is accepted): *)
(* undefined - nothing has been read yet (empty stream) *)
(* first_comment_intro - inside a leading comment that runs to EOF *)
(* with no trailing newline (comment-only stream) *)
(* first_bom - BOM or first-line comment (its newline seen), *)
(* no assignment yet *)
(* value_outro - the semicolon of the last assignment consumed *)
(* In all four cases struct_nesting_level must be zero. *)

(* Events emitted to the validator callbacks (bvnr_event_t): *)
(* *)
(* ev_stream_start - once at the very start of bvn_lex_run *)
(* ev_stream_end - once at the very end of bvn_lex_run, *)
(* after the last value's terminator is consumed *)
(* ev_assignment_start - on the identifier token of an assignment *)
(* ev_struct_start - on "{" *)
(* ev_struct_end - on "}" *)
(* ev_array_row_start - on "[" opening a bracket row *)
(* ev_array_row_end - on "]" closing a bracket row *)
(* ev_array_dim_start - on "/" between rows; fired immediately *)
(* when "/" is consumed, before the next *)
(* "[" opens the next row *)
(* ev_octet_stream_start - on the NUL byte that enters binary mode *)
(* ev_octet_stream_end - on the 0x00 tag ending an octet stream *)
(* *)
(* For every typed value (including values whose type annotation is *)
(* synthesised by the validator - see §4) the following sequence *)
(* is emitted before ev_data: *)
(* *)
(* ev_type_annotation_start - once, before params *)
(* ev_type_annotation_type_family - the family name *)
(* ev_type_annotation_type_family_parameter - zero to three *)
(* times, once per present parameter (width, *)
(* base, unit), in that order *)
(* ev_type_annotation_end - once, after params *)
(* ev_data - the value token *)
(* *)
(* ev_type_annotation_start is emitted to on_unverified first *)
(* (raw bytes only), then to on_verified (with value_type and *)
(* value_unit filled in). All other type-annotation events and *)
(* ev_data are emitted to both callbacks simultaneously. *)
(* For bool keywords (true/false/on/off) the same default type- *)
(* annotation sequence is emitted, synthesising <bool> when no *)
(* explicit annotation was given (no width/base/unit params). *)
(* *)
(* ev_data is also emitted for null values, symbols, references, *)
(* and octet-stream chunks without a preceding type-annotation *)

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(*) sequence. Struct values do NOT produce ev_data; they are *)
(*) signaled by ev_struct_start and ev_struct_end. *)

stream      = [utf8-bom] , ws , [version-directive , ws] ,
              {assignment , ws} ;

(*) UTF-8 BOM: legal only as the very first three bytes of the stream. *)
utf8-bom    = "\xEF\xBB\xBF" ;

(*) Version directive (spec 1.1): an OPTIONAL leading comment of the *)
(*) form "#!bovnar <major>.<minor>" declaring the spec version the *)
(*) document targets. It is recognised only as the very first comment *)
(*) (after the optional BOM and whitespace); a "#!bovnar ..." on any *)
(*) later line is an ordinary comment. Lexically it IS a comment, so a *)
(*) spec-1.0 reader skips it transparently – the declaration is purely *)
(*) additive. A 1.1+ reader parses it, exposes the version, and (in *)
(*) strict mode) rejects a version it does not support. Each component *)
(*) is a decimal integer with no leading zero (a bare "0" excepted). *)
(*) A malformed version after the "!bovnar" prefix is *)
(*) error_invalid_spec_version. *)
version-directive = "#" , "!bovnar" , ws-mandatory ,
                   version-int , "." , version-int ,
                   {ws-char} , (" \x0D" | " \x0A" | (* eof *)) ;
(*) version-int: DIGIT , {DIGIT} with no leading zero (a bare "0" is the *)
(*) only form starting with "0"); each component must fit in a uint16. *)
version-int = DIGIT , {DIGIT} ;

(*) Whitespace: blank characters and comments, freely intermixed. *)
ws          = {ws-char | comment} ;

(*) At least one whitespace character, followed by optional further *)
(*) whitespace and comments. Used as the mandatory separator before *)
(*) an inline unit suffix. *)
ws-mandatory = ws-char , {ws-char | comment} ;
ws-char      = " \x09"          (* HT *)
              | " \x0A"          (* LF *)
              | " \x0B"          (* VT *)
              | " \x0C"          (* FF *)
              | " \x0D"          (* CR *)
              | " \x20" ;       (* SP *)

(*) Comment: "#" through the next LF, CR, or end of input. *)
(*) Control bytes 0x00–0x08, 0x0E–0x1F, and 0x7F are hard errors *)
(*) inside a comment body. *)
comment      = "#" , {comment-char} , (" \x0D" | " \x0A" | (* eof *)) ;
comment-char = " \x09"          (* HT *)
              | " \x0B" | " \x0C" (* VT, FF *)
              | printable-byte   (* 0x20–0x7E *)
              | high-byte ;      (* 0x80–0xFF *)

(*) _____ *)
(*) 2. Assignments *)
(*) _____ *)

(*) The leading "." is a syntactic sigil; it is not stored in the key. *)
(*) At least one id-start character after "." is required; *)
(*) ".=" is error_empty_identifier. *)
(*) Comments are permitted between the last key character and "=". *)

assignment   = "." , key , ws , "=" , ws , value , ";" ;

key          = id-start , {id-body-char} ;

(*) First character of an identifier, symbol, or reference segment: *)
(*) A–Z, a–z, "_", and UTF-8 leader bytes 0xC3–0xF4. *)
(*) 0xC2 is explicitly rejected (ACT_NONE) at all token-start *)

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(* positions, so code points U+0080–U+00BF cannot begin these tokens. *)
id-start      = ALPHA | "_" | utf8-leader ;

(* Body characters of an identifier: *)
(* Accepted: "+", "-", digits, alpha, "_", *)
(* UTF-8 leader bytes 0xC3–0xF4, UTF-8 continuation bytes 0x80–0xBF. *)
(* Hard-error bytes: 0x00–0x1F, 0x7F, 0xC2, and ASCII punctuation: *)
(* " # , . / ; < > [ ] { } = ! $ % & ' ( ) * : ? @ \ ^ ` | ~ *)
(* "=" terminates the body (ACT_value_intro). *)
(* Whitespace terminates the body (ACT_to_identifier_outro), after *)
(* which only "=" is accepted (with comments permitted between). *)
id-body-char = "+" | "-"
              | DIGIT
              | ALPHA
              | "_"
              | utf8-leader      (* 0xC3–0xF4 *)
              | utf8-continuation ; (* 0x80–0xBF *)

(* ----- *)
(* 3. Values *)
(* ----- *)

(* A value is the right-hand side of an assignment or an element *)
(* inside an array row; the optional type annotation applies in both *)
(* contexts. *)
(* *)
(* Typed null elements: in array context, a type annotation not *)
(* followed by any raw value token (i.e. the next byte is "," "]" *)
(* or ";") is a null value carrying the annotated type. This is *)
(* handled by ACT_type_null_then_new_array_value, *)
(* ACT_type_null_then_array_outro, and ACT_type_null_then_value_outro *)
(* from the type_outro state. *)

value      = [type-annotation , ws] , raw-value ;

raw-value  = null-value
            | bool-value
            | special-number
            | datetime-literal      (* spec 1.1 *)
            | ( number | string ) , [ ws-mandatory , inline-unit ]
            | symbol
            | reference
            | array
            | struct
            | octet-stream ;

(* datetime-literal (spec 1.1): an ISO-8601 UTC timestamp accepted in *)
(* place of the integer epoch-seconds carrier. The lexer accumulates the *)
(* shape (dtlit_* states, entered from a digit run followed by "-"); the *)
(* validator strictly range-checks the fields and converts the UTC civil *)
(* instant to the integer carrier on the declared epoch (unix when a bare *)
(* literal infers <datetime:64,unix>). Whole-second carrier: a frac- *)
(* tional part (any digit count) does not change the value but is kept *)
(* verbatim and re-emitted so the literal round-trips. A "+HH:MM" / *)
(* "-HH:MM" offset is folded to true UTC before conversion. *)
(* The atomic GNSS epochs (gps/galileo/glonass/beidou) reject a literal. *)
datetime-literal = year , "-", month , "-", day ,
                  [ "T" , hour , ":" , minute , ":" , second ,
                    [ "." , dt-fraction ] , [ dt-zone ] ] ;
dt-fraction      = digit , { digit } ;      (* kept verbatim; carrier is whole seconds *)
dt-zone          = "Z" | ( "+" | "-" ) , hour , ":" , minute ; (* tz offset *)
year             = digit , digit , digit , digit ;
month            = digit , digit ;          (* 01..12 *)
day              = digit , digit ;          (* 01..28/29/30/31 per month *)
hour             = digit , digit ;          (* 00..23 *)
minute           = digit , digit ;          (* 00..59 *)

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second      = digit , digit ;    (* 00..59 *)

(* null-value: either the reserved keyword "null" or the empty *)
(* production. *)
(* At assignment level: ".foo = ;" and ".foo = null;" are equivalent. *)
(* In array context: a leading or trailing comma, two consecutive *)
(* commas, a bare "null", or a type annotation immediately followed *)
(* by ",", " / "]" all denote a null element. *)
null-value  = "null" | (* empty *) ;

(* bool-value: a reserved keyword in value position. "on" is an *)
(* alias for "true" and "off" for "false"; the validator collapses *)
(* all four spellings to a vt_bool value (canonical text "true" / *)
(* "false") and synthesises a <bool> type annotation when none is *)
(* given (see §4). An explicit <bool> annotation accepts only these *)
(* keywords as its value. *)
(* *)
(* Lexically these keywords – together with "null" – are ordinary *)
(* symbol tokens (§7); the validator reserves the exact words and *)
(* reclassifies them. A longer word that merely starts with one *)
(* (e.g. "truthy", "nullable") remains a plain symbol. *)
bool-value  = "true" | "false" | "on" | "off" ;

(* ----- *)
(* 4. Type annotations *)
(* ----- *)

(* Lexer: a keyword state machine (tf_* states) recognises the eight *)
(* family names via six keyword paths (float_fix / float_dec share *)
(* the "float" prefix; see §4 below). After the family keyword the *)
(* raw bytes of the ":"... *)
(* parameter string are accumulated into type_data by copy_type_byte. *)
(* *)
(* Validator: bv_parse_type_annotation parses the accumulated string *)
(* and the validator emits the structured type-annotation event *)
(* sequence described in §1. *)
(* *)
(* Default synthesis: when a number or string value carries no *)
(* explicit type annotation (value_type is still vt_plain), the *)
(* validator synthesises a default before emitting ev_data: *)
(* strings / special-numbers → <utf8> / <float:64,_10,no_unit> *)
(* floats (has "." or "e"/"E") → <float:64,_10,no_unit> *)
(* negative integers          → <sint:64,_10,no_unit> *)
(* plain integers             → <uint:64,_10,no_unit> *)
(* bool keywords (true/false/on/off) → <bool> *)
(* The full type-annotation event sequence is always emitted for *)
(* number and string values, whether annotated explicitly or not. *)

type-annotation = "<" , ws , type-spec , ws , ">" ;

type-spec      = param-type ;

(* Types with an optional parameter list after ":". *)
(* Parameters are identified by class and may appear in any order; *)
(* at most one of each class (width, base, q, unit) is permitted. *)
(* *)
(* "utf8" goes to type_body_outro after the keyword; the lexer *)
(* accepts ":" and the parameter bytes, but utf8 is a parameterless *)
(* family – any width, base, q, or unit parameter is rejected by *)
(* bv_parse_type_annotation as error_illegal_value_type (see bool). *)
(* *)
(* "float_fix" and "float_dec" are not separate keyword paths in the *)
(* lexer state table. The lexer fires ACT_tf_float_done after the *)
(* shared "float" prefix (states tf_f → tf_fl → tf_flo → tf_floa *)
(* → ACT_tf_float_done), which stores "float" in type_data and *)
(* transitions to type_body_outro. The subsequent bytes "_fix" or

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(* "_dec" are accumulated via copy_type_byte. The final string *)
(* ("float", "float_fix", or "float_dec") is dispatched in *)
(* bv_n_parse_type_annotation. *)
(* "bool" and "utf8" are parameterless families: any width, base, q, *)
(* or unit parameter is error_illegal_value_type. bool's only valid *)
(* values are the bool keywords (see bool-value, §3). *)
(* "datetime" (spec 1.1) is a timestamp: a signed integer seconds count *)
(* validated like sint, with one epoch-name parameter (unix|tai|gps|mjd| *)
(* ntp|galileo|glonass|y2000|beidou; default unix). A numeric base, q, or *)
(* unit parameter is error_illegal_value_type. Available only when the *)
(* document declares "#!bovnar 1.1" or newer. *)
param-type = ("uint" | "sint" | "float" | "float_fix" | "float_dec" | "utf8"
              | "bool" | "datetime")
            , [ws , ":" , ws , type-param-list] ;

(* The parameter list is accumulated byte-by-byte into type_data by *)
(* the copy_type_byte / type_body_outro states, and later re-parsed *)
(* by bv_n_parse_type_annotation. *)
(*)
(*)
(* Bytes accepted in the accumulated string: *)
(*)
(*) A-Z, a-z, 0-9, "*", "+", ",", ".", "/", "-", ":", "^", "_" *)
(*)
(*) UTF-8 continuation bytes 0x80-0xBF *)
(*)
(*) UTF-8 leader bytes 0xC2-0xF4 (0xC2 IS accepted here, unlike *)
(*)
(*) identifier / symbol / reference token starts) *)
(*)
(*) Whitespace is accepted but not accumulated; it transitions to *)
(*)
(*) type_body_outro, which accepts the same bytes as copy_type_byte *)
(*)
(*) plus "#" (0x23, entering comment_intro), while treating whitespace *)
(*)
(*) as ignored rather than re-triggering the outro transition. *)
(*)
(*) ">" always terminates the annotation. *)
type-param-list = type-param , {ws , "," , ws , type-param} ;

type-param = width-param          (* e.g. 32, 64 *)
            | base-param          (* e.g. _16, _64, _85 *)
            | q-param             (* e.g. q8, q16 – float_fix only *)
            | unit-param ;        (* e.g. m~s, Gi~B, m^2 *)

(*)
(*) Width: one or more decimal digits. 0 means "no width restriction".*)
(*)
(*) float: 0, 16, or any multiple of 32 up to BVN_FLOAT_MAX_PREC.*)
(*)
(*) float_fix, *)
(*)
(*) float_dec: 0, 16, 32, 64, 128, 256. *)
width-param = DIGIT , {DIGIT} ;

(*)
(*) Base: underscore followed by one or more decimal digits. *)
(*)
(*) Valid values: 2-62, 64 (standard Base64), 85 (Ascii85). *)
(*)
(*) Bases 64 and 85 are uint-only: their alphabets use '+'/'-' as *)
(*)
(*) digits, leaving no sign character, so sint+64/85 is illegal. *)
(*)
(*) "float" accepts only base 10 (or absent, → 10) or base 16; *)
(*)
(*) other bases → error_illegal_value_type. *)
(*)
(*) "float_fix" and "float_dec" reject the base parameter entirely: *)
(*)
(*) any base-param → error_illegal_value_type. *)
(*)
(*) A non-decimal base with uint/sint should use a quoted string value; *)
(*)
(*) a bare literal is not rejected by the validator but will be parsed *)
(*)
(*) as a symbol token, causing a type/value mismatch instead. *)
base-param = "_" , DIGIT , {DIGIT} ;

(*)
(*) Q-parameter: lowercase "q" followed by one or more decimal digits. *)
(*)
(*) Specifies the number of fractional bits for the float_fix Q-format. *)
(*)
(*) Value = raw_signed_integer × 2(-Q). *)
(*)
(*) Valid only for "float_fix"; rejected for all other families. *)
(*)
(*) Valid range: 0 ≤ Q < effective_width (width 0 → effective 64). *)
(*)
(*) Q is stored in value_type_spec_t.base and retrieved via *)
(*)
(*) bv_n_effective_q; bv_n_effective_base always returns 10 for *)
(*)
(*) float_fix and float_dec. *)
q-param = "q" , DIGIT , {DIGIT} ;

(*)
(*) Unit parameter: everything up to the next ",", " or ">". *)
(*)
(*) Parsed against the unit table by bv_n_parse_unit. *)

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(* "no_unit" is a reserved keyword meaning explicitly dimensionless. *)
(* Valid for: uint, sint, float, float_fix, float_dec. *)
unit-param = unit-char , {unit-char} ;
unit-char  = (* any byte accepted by copy_type_byte that is not *)
              (* ",", (0x2C) or ">" (0x3E): *)
              (* A-Z, a-z, 0-9, "*", ".", "/", "-", ":", "^", "_", *)
              (* "~", "$", "(", ")", or a UTF-8 multi-byte byte *)
              (* (0x80-0xF4) *)
              ;

(* — Unit sub-grammar (semantic, enforced by bvn_parse_unit) — *)
(* *)
(* resolved-unit = "no_unit" | unit-expr ; *)
(* unit-expr    = unit-factor , {unit-sep , unit-factor} ; *)
(* unit-factor   = unit-component | "(" , unit-expr , ")" ; *)
(* *)
(* unit-sep      = "*" (* product – U+002A *) *)
(*               | "." (* product – U+00B7 *) *)
(*               | "/" ; (* division *) *)
(* ". " is the middle dot, encoded as 0xC2 0xB7 (UTF-8 for U+00B7). *)
(* "*" and "." are semantically equivalent (multiplication). *)
(* *)
(* Semantics of "/": the first "/" switches every subsequent factor *)
(* into the denominator (its stored exponents are negated). A "(...)" *)
(* group is a sub-expression evaluated independently; when it falls in *)
(* the denominator its net exponents are negated as a whole, so *)
(* "k-g/(m·s²)" = kg·m-1·s-2 and "(k-g/m)·s²" = kg·m-1·s². Additional *)
(* "/" separators at one level never toggle back to the numerator. *)
(* An explicit separator is required before a group ("m(s)", not *)
(* "m(s)"); a group is not (yet) followed by its own exponent; paren *)
(* nesting is bounded at 16. *)
(* *)
(* Empty components or groups (e.g. "m//s", "()", "m*s"), unmatched *)
(* parentheses, and more than BVN_MAX_UNIT_COMPONENTS = 8 total *)
(* components per unit string all produce error_unit_illegal. *)
(* *)
(* unit-component = [prefix "~"] base-unit [unit-exponent] ; *)
(* *)
(* When a prefix is present the separator "~" is mandatory. *)
(* A bare base-unit with no prefix requires no separator. *)
(* *)
(* si-prefix = "Q"|"R"|"Y"|"Z"|"E"|"P"|"T"|"G"|"M"|"k"|"h"|"da" *)
(*            | "d"|"c"|"m"|"μ"|"n"|"p"|"f"|"a"|"z"|"y"|"r"|"q" ; *)
(* (μ = U+00B5: 0xC2 0xB5) *)
(* *)
(* iec-prefix = "Ki"|"Mi"|"Gi"|"Ti"|"Pi"|"Ei"|"Zi"|"Yi"|"Ri"|"Qi" ; *)
(* Ri (2^90) and Qi (2^100) were added in IEC 80000-13:2022 ed. 2. *)
(* *)
(* base-unit = "b"|"B"|"s"|"m"|"g"|"A"|"K"|"mol"|"cd"|"Hz"|"N" *)
(*            | "Pa"|"J"|"W"|"V"|"Ω"|"F"|"C"|"S"|"Wb"|"T"|"H" *)
(*            | "lm"|"lx"|"Bq"|"Gy"|"Sv"|"kat"|"rad"|"sr" *)
(*            | "L"|"l"|"min"|"h"|"d"|"wk"|"yr" *)
(*            | "°C"|"°"|"degC"|"degrC" *)
(*            | "degrees"|"degree"|"degr"|"deg" *)
(*            | "t"|"bar"|"eV"|"Da"|"au"|"ha" ; *)
(* (Ω = U+2126: 0xE2 0x84 0xA6; ° = U+00B0: 0xC2 0xB0) *)
(* *)
(* unit-exponent = [exp-sign] exp-digit *)
(*               | "^" ["-"|"+"] ASCII-digit ; *)
(* *)
(* exp-sign = "+" (* U+207A: 0xE2 0x81 0xBA, positive / no-op *) *)
(*           | "-" ; (* U+207B: 0xE2 0x81 0xBB, negate exponent *) *)
(* *)
(* exp-digit = "1"|"2"|"3"|"4"|"5"|"6"|"7"|"8"|"9" ; *)
(* (U+00B9 / 00B2 / 00B3 / 2074–2079) *)
(* *)
(* "+" is accepted but has no effect (same as absent). *)
(* The ASCII form "[+~]?[1-9]" maps to the same internal constants *)

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(* as the Unicode superscript form. "^0" is not a valid exponent. *)

(* ----- *)
(* 5. Numbers *)
(* ----- *)

(* The lexer is digit-agnostic: any ASCII digit 0-9 is accepted in a *)
(* bare number literal without base validation. Digit validity against *)
(* a non-decimal base is checked later by bvn_acc_parse_number. *)
(* Leading zeros are not rejected by the lexer ("007" is valid). *)
(* Only "e"/"E" are accepted as exponent markers in bare literals. *)
(* "p"/"P" (binary exponent) is recognised by the validator inside a *)
(* quoted-string number value of a base-16 float, e.g. *)
(* <float:64,_16> "1.8p+2"; there "e"/"E" are hex mantissa digits, so *)
(* the binary exponent must use "p"/"P" (see spec §6.3). It is the *)
(* only context in which "p"/"P" is reachable. *)

number      = ["-"] , (int-led | dot-led) , [dec-exponent] ;

(* Integer-led: one or more digits with an optional fractional part. *)
(* A trailing dot ("123.") without fractional digits is valid. *)
int-led     = DIGIT , {DIGIT} , [ "." , {DIGIT} ] ;

(* Dot-led: at least one digit after the dot is required. *)
(* ".5" and "-.5" are valid; a lone "." is a hard error. *)
dot-led     = "." , DIGIT , {DIGIT} ;

dec-exponent = ("e" | "E") , ["+" | "-"] , DIGIT , {DIGIT} ;

(* Special IEEE-754 values are bare reserved keywords: nan, inf, and *)
(* ninf (negative infinity). Like null / true / false / on / off they *)
(* are lexed as ordinary symbols and reclassified to numeric special *)
(* values by the validator; a bare word that is not one of these exact *)
(* spellings stays an ordinary symbol (e.g. "infinity", "nans"). *)
(* There is no sigil and no inline-unit suffix – a unit is supplied *)
(* via the type annotation, e.g. <float:64,m/s> inf. *)
special-number = "nan" | "inf" | "ninf" ;

(* ----- *)
(* 6. Strings *)
(* ----- *)

(* Two or more adjacent quoted literals with only whitespace/comments *)
(* between them are concatenated into a single string token by the *)
(* lexer (string_outro re-enters string_intro on ''). The combined *)
(* byte length is bounded by max_string_length (default UINT16_MAX). *)

string      = string-literal , {ws , string-literal} ;

string-literal = '"' , {string-char} , '"' ;

string-char = safe-string-byte
            | escape-seq ;

(* safe-string-byte: whitespace 0x09-0x0D, 0x20-0x7E except '"' (0x22) *)
(* and '\' (0x5C), plus high bytes 0x80-0xFF. *)
(* Control bytes 0x00-0x08, 0x0E-0x1F, and DEL (0x7F) are hard *)
(* errors even inside strings. *)
safe-string-byte = (* byte in [0x09,0x0D] | [0x20,0x21] | [0x23,0x5B] *)
                  (* | [0x5D,0x7E] | [0x80,0xFF] *) ;

(* Recognised escape sequences (from bvn_escape_lut): *)
escape-seq = "\" , ( "t" (* \t = 0x09 HT *)
                    | "n" (* \n = 0x0A LF *)
                    | "v" (* \v = 0x0B VT *)

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| "f"      (* \f = 0x0C FF *)
| "r"      (* \r = 0x0D CR *)
| "'"      (* \" *)
| "\"      (* \\ *)
| byte-escape      (* spec 1.1, see below *)
| unicode-escape ) ; (* spec 1.1, see below *)
(* All other bytes after "\" are error_illegal_escape_sequence. *)
(* spec 1.1 only – available when the document declares "#!bovnar 1.1" *)
(* or newer; in a 1.0/unversioned document "x"/"u" after "\" stay *)
(* error_illegal_escape_sequence. *)
byte-escape      = "x" , hex-digit , hex-digit ;      (* one byte 0xHH *)
unicode-escape = "u" , "{" , hex-digit , {hex-digit} , "}" ;
(* 1..6 hex digits naming a Unicode scalar; surrogates (D800..DFFF) and *)
(* values > 10FFFF are error_invalid_codepoint, UTF-8 encoded otherwise. *)
(* A \x byte must leave the whole string valid UTF-8 (else *)
(* error_invalid_utf8_byte). *)
hex-digit      = DIGIT | "a".."f" | "A".."F" ;

(* ----- *)
(* 6.5 Inline unit suffix *)
(* ----- *)

(* An inline-unit may follow a scalar number or string value (but not *)
(* a special-number keyword – give its unit via the type annotation). *)
(* At least one whitespace character (ws-mandatory) is the sole *)
(* required separator between the value literal and the unit. *)
(* Inline units are forbidden inside array elements (enforced at the *)
(* action level; not expressible in context-free EBNF). *)
(* *)
(* INVALID (no whitespace before the unit): *)
(* .bad = 9.81m/s;      (* no separator at all – hard error *) *)
(* *)
(* Examples: *)
(* .a = 9.81 m/s;      (* space separator *) *)
(* .b = <float:64> 1.5 k~m; (* with type annotation *) *)
(* .c = <float:64,s> inf; (* special-number unit via annotation *) *)
(* .d = 1.5e3 k~J;      (* number with exponent *) *)
(* .e = "FF" m;      (* string literal with unit *) *)
(* *)
(* State-machine flow: *)
(* 1. Whitespace from a number-body state fires ACT_to_number_outro *)
(* → number_outro. Strings enter string_outro_nosp after the *)
(* closing "\"; whitespace transitions to string_outro. *)
(* Special numbers (nan, inf, ninf) are lexed as symbols and *)
(* take no inline unit (give the unit via the type annotation). *)
(* 2. From number_outro / string_outro (whitespace already seen): *)
(* alpha / "\" / UTF-8 leader → ACT_inline_unit_intro *)
(* 3. ACT_inline_unit_intro pushes the first byte and enters *)
(* inline_unit_body directly. *)
(* 4. inline_unit_body accumulates unit characters (including ":" *)
(* as a valid mid-unit byte) until whitespace (→ *)
(* inline_unit_outro) or ";" (→ value_outro via finalize). *)
(* *)
(* Inside an array element (in_array_element flag set), *)
(* ACT_inline_unit_intro immediately rejects with *)
(* error_unexpected_input_byte. *)
(* *)
(* Semantic rule (enforced by validator): *)
(* If a type-annotation unit was also given, the inline unit must *)
(* parse to the identical value_unit_t; mismatch → error_unit_mismatch. *)

inline-unit      = inline-unit-start , { inline-unit-char } ;

(* A currency component carries a mandatory "$" sigil (spec 1.0): "$USD", *)
(* "$BTC", or prefixed "<prefix>~$EUR". Bare codes are no longer currencies, *)
(* so a currency code can never collide with a physical-unit symbol. *)

```

```

(* "(" lets an inline unit begin with a parenthesised group, e.g. "(m/s)/s", *)
(* so the inline form accepts the same unit grammar as a type-annotation unit. *)
inline-unit-start = ALPHA | "_" | "$" | "%" | "(" | unit-lead-byte ;

inline-unit-char  = ALPHA | DIGIT | "_" | "$" | "%" | "+" | "-" | "." | "/" | ":" | "^" |
"*" | "~"
                  | "(" | ")" | unit-lead-byte | utf8-continuation ;

(* unit-lead-byte: the UTF-8 lead-byte range accepted inside a unit – the *)
(* full 0xC2-0xF4, BROADER than utf8-leader (0xC3-0xF4). Unlike identifier, *)
(* symbol, and reference token starts (where 0xC2 is rejected, see §12), a *)
(* unit may begin with or contain a 0xC2-led code point: the μ prefix *)
(* (U+00B5 = 0xC2 0xB5), the ° / °C units (U+00B0 = 0xC2 0xB0), and the *)
(* "." product separator (U+00B7 = 0xC2 0xB7). The lexer maps the whole *)
(* 0xC2-0xF4 range to ACT_inline_unit_intro / ACT_copy_inline_unit_byte, *)
(* matching copy_type_byte in a type-annotation unit body. *)
unit-lead-byte   = (* byte in [0xC2, 0xF4] *) ;

(* ----- *)
(* 7. Symbols (unquoted bare-word values in value position) *)
(* ----- *)

(* A symbol starts with id-start and continues with sym-body-char. *)
(* Differences from the identifier body action table: *)
(* "=" is a hard error (not a state transition) *)
(* ",", " / "]" / ";" terminate the symbol cleanly (not errors) *)
(* "." is a hard error (same as in identifier body) *)

(* Reserved keywords: a symbol token whose text is exactly "null", *)
(* "true", "false", "on", or "off" is reclassified by the validator *)
(* (see null-value / bool-value, §3) and is NOT delivered as a symbol. *)
(* Any other bare word – including one that merely begins with a *)
(* keyword, like "ontology" or "nullish" – remains a symbol. *)
symbol          = id-start , {sym-body-char} ;

(* sym-body-char spans the same byte range as id-body-char; the *)
(* difference is purely in the termination actions fired for *)
(* ",", " (new_array_value), "]" (array_outro), and ";" (value_outro). *)
sym-body-char    = id-body-char ;

(* ----- *)
(* 8. References *)
(* ----- *)

(* The "&" sigil introduces a dotted path to another key. *)
(* The stored string starts with "." and includes all dots: *)
(* "&.foo.bar" stores ".foo.bar". *)
(* At least one segment is required; "&" alone or "&." without a *)
(* following id-start character is an error. *)
(* A reference is stored UNRESOLVED – the path is a string token; the *)
(* library never dereferences it, so the target need not exist and *)
(* cycles are not detected. Resolution is the application's job. *)

reference       = "&" , ref-segment , {ref-segment | ref-index} ;
ref-segment    = "." , id-start , {ref-body-char} ;

(* ref-index (spec 1.1): array indexing in a reference path, e.g. *)
(* &.matrix[0][1]. The bytes are captured verbatim into the path *)
(* string (references are stored unresolved); bvn_dom_lookup *)
(* interprets them at the DOM layer. Gated on a "#!bovnar 1.1" *)
(* declaration – in a 1.0/unversioned document "[" in a reference is *)
(* error_unexpected_input_byte. The reference_index sub-state *)
(* distinguishes a "]" that closes an index from one that closes an *)
(* enclosing array (which ends the reference). *)
ref-index      = "[" , DIGIT , {DIGIT} , "]" ;

```

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(* ref-body-char: same byte set as sym-body-char, except that *)
(* "." (0x2E) is ACT_copy_reference_dot, which pushes the dot into *)
(* the token buffer and advances to reference_segment_intro for the *)
(* next segment's id-start character. *)
ref-body-char = sym-body-char | "." ;

(* ----- *)
(* 9. Arrays *)
(* ----- *)

(* An array is one or more bracket-enclosed rows separated by "/": *)
(* .a = [1, 2, 3]/[4, 5, 6]; *)
(* "/" emits ev_array_dim_start; the next "[" opens the next row. *)
(* Only whitespace/comments between "/" and "[" are permitted. *)
(* The former adjacent-row syntax "[1,2][3,4]" and the comma-between- *)
(* rows form "[1,2],[3,4]" are hard errors at the assignment level. *)
(* array_outro no longer accepts "[" directly; the only continuation *)
(* for a further row is "/". *)
(* Nested arrays are valid element values inside a row: *)
(* [[1,2],[3,4]] is a one-row outer array whose two elements are *)
(* the inner arrays [1,2] and [3,4]. *)
(* Leading and trailing commas produce null elements: *)
(* [1,2,] → four elements: null, 1, 2, null. *)
(* Semantic constraints (enforced by the lexer/validator): *)
(* a) The "/"-separated dimension rows of a SINGLE array must all *)
(* have the same element count; a mismatch is *)
(* error_array_row_size_mismatch (code 2). The lexer checks each *)
(* row as it closes, so the error fires at the earliest byte (the *)
(* overshooting element, or the closing "]" of a short row). *)
(* The per-bracket-pair context save/restore mechanism keeps this *)
(* check scoped to one array instance: a "/"-row width never *)
(* leaks across a "," boundary or into a sibling branch. *)
(* Beyond /-row consistency, array elements must be HOMOGENEOUS *)
(* (spec 1.0, enforced above the grammar): same kind, same *)
(* dimension for bare scalar arrays, and rectangular sibling *)
(* sub-arrays. Examples: *)
(* [1,2,3]/[4,5] → error (one array's /-rows: 3 vs 2) *)
(* [[1,2]/[3,4,5]] → error (inner /-array's rows: 2 vs 3) *)
(* [[1,2],[3,4,5]] → error (ragged sibling sub-arrays) *)
(* [[1,2]/[3,4], [5,6]/[7,8]] → accepted (same-shape blocks) *)
(* b) Array nesting depth ≤ max_array_nesting (default 64, cap 255).*)

array = array-row , {"/" , ws , array-row} ;

array-row = "[" , ws , [row-content] , ws , "]" ;

(* row-content is OPTIONAL: "[" (nothing between the brackets) is an *)
(* EMPTY row – zero elements, no ev_data. It is distinct from "[null]" *)
(* (one null element) and "[,]" (two null elements). When present, *)
(* row-content has at least one element position; leading / trailing / *)
(* consecutive commas each expand to a null-value element. An empty row *)
(* has width 0, a valid row width for /-dimension consistency. *)
row-content = array-elem , {ws , "," , ws , array-elem} ;

(* array-elem has the same structure as a top-level value. *)
array-elem = [type-annotation , ws] , raw-value ;

(* ----- *)
(* 10. Structs (nested key-value scopes) *)
(* ----- *)

```

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(* A struct "{" ... "}" contains zero or more assignments. "{}" is an *)
(* empty struct (zero members). After "}" the struct behaves like any *)
(* completed value: followed by ";" at the top level, or "," / "]" in *)
(* array context. *)
(*)
(* Keys must be UNIQUE within one scope (a struct, or the top-level *)
(* document); a repeat is error_duplicate_struct_key (enforced above *)
(* the grammar). The same key in different scopes is unrelated. *)
(* Nesting depth: struct_nesting_level ≤ max_struct_nesting *)
(* (default 64 when the flag is 0; hard cap 255). *)
(* An unmatched "}" is error_illegal_struct_close. *)

struct      = "{" , ws , {assignment , ws} , "}" ;

(*)
(*) ----- (*)
(* 11. Octet streams (binary-framed escape from the text layer) *)
(*) ----- (*)

(* A NUL byte (0x00) appearing where a value is expected switches from*)
(* text mode to binary chunk mode. The UTF-8 stream validator is *)
(* suspended for the duration. *)
(*)
(*) Binary protocol (bvn_read_octet_stream): *)
(*) Tag 0x01 → chunk follows (2-byte LE length + data bytes) *)
(*) Tag 0x00 → end of stream *)
(*) Any other tag → error_octet_stream_out_of_sync *)
(*)
(*) After ev_octet_stream_end the lexer enters octet_stream_outro, *)
(*) which accepts the same terminators as any other completed value. *)

octet-stream = "\x00" , {os-chunk} , os-end ;

os-chunk     = "\x01" , os-length , os-data ;
os-end       = "\x00" ;

(* 16-bit little-endian length; 0x0000 encodes 65536 bytes. *)
os-length    = byte , byte ; (* little-endian uint16; 0x0000 → 65536 *)

os-data      = {byte} ; (* exactly os-length bytes; not CF *)

(*)
(*) ----- (*)
(* 12. Lexical primitives *)
(*) ----- (*)

DIGIT        = "0" | "1" | "2" | "3" | "4"
              | "5" | "6" | "7" | "8" | "9" ;

ALPHA        = "A" | "B" | "C" | "D" | "E" | "F" | "G" | "H" | "I" | "J" | "K" | "L" | "M"
              | "N" | "O" | "P" | "Q" | "R" | "S" | "T" | "U" | "V" | "W" | "X" | "Y" | "Z"
              | "a" | "b" | "c" | "d" | "e" | "f" | "g" | "h" | "i" | "j" | "k" | "l" | "m"
              | "n" | "o" | "p" | "q" | "r" | "s" | "t" | "u" | "v" | "w" | "x" | "y" | "z" ;

(* utf8-leader here denotes the range used at identifier, symbol, and *)
(* reference token-start and -body positions: 0xC3–0xF4. *)
(* 0xC2 is the leader for U+0080–U+00BF. It is explicitly overridden *)
(* to ACT_NONE at every identifier, symbol, and reference boundary, *)
(* so code points U+0080–U+00BF cannot start or appear in those *)
(* tokens. 0xC2 IS accepted inside quoted strings and type parameter *)
(* bodies (copy_type_byte), where the full range 0xC2–0xF4 applies. *)
utf8-leader  = (* byte in [0xC3, 0xDF] | [0xE0, 0xEF] | [0xF0, 0xF4] *) ;

utf8-continuation = (* byte in [0x80, 0xBF] *) ;

(* Printable ASCII: 0x20–0x7E *)

```

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printable-byte = (* byte in [0x20, 0x7E] *) ;

(* High byte: any byte 0x80–0xFF. 0xC0, 0xC1, 0xF5–0xFF are *)
(* structurally in this range but rejected by the parallel UTF-8 *)
(* validator (error_invalid_utf8_byte). *)
high-byte      = utf8-continuation | utf8-leader
                | (* 0xC0 | 0xC1 | 0xF5–0xFF – rejected by UTF-8 check *) ;

byte           = (* any single byte 0x00–0xFF *) ;

(* _____ *)
(* 13. Constraints not expressible in context-free EBNF *)
(* _____ *)

(*)
a) UTF-8 validity
Every byte in the text layer (outside octet-stream regions) must
form valid UTF-8 when fed through bvn_utf8_feed. The UTF-8
validator runs in parallel with the action state machine; a
non-UTF-8 byte causes error_invalid_utf8_byte regardless of which
lexer state the parser is in. Overlong encodings and surrogate
halves are rejected.

b) BOM uniqueness
The UTF-8 BOM (EF BB BF) is legal only at byte offset 0.
If the stream begins with "#" instead (a first-line comment), the
special first_comment_* state machine watches the comment body for
the exact byte sequence 0xEF 0xBB 0xBF and rejects it as
error_invalid_byte_order_mark. This detection applies only while
the first_comment_* states are active (i.e. until the first
newline); subsequent regular comments (comment_intro state) have
no BOM detection, so U+FEFF encoded as 0xEF 0xBB 0xBF appearing
in any later comment body or string literal is valid UTF-8 and is
accepted.

c) 0xC2 at token boundaries
The byte 0xC2 (UTF-8 leader for U+0080–U+00BF) is mapped to
ACT_NONE at identifier, symbol, and reference token-start and
token-body positions, forbidding code points U+0080–U+00BF in
those tokens. 0xC2 is accepted inside quoted strings and in type
parameter bodies (copy_type_byte / type_body_outro).

d) Special-number keywords
The special floats nan / inf / ninf are bare reserved keywords (no
sigil). They are lexed as symbols and reclassified to numeric
special values by the validator, like null / true / false / on /
off; any other bare word (e.g. "infinity", "nans") stays a symbol.
They take no inline unit – a unit is supplied via the annotation.

e) String concatenation
Adjacent string literals with only whitespace/comments between
them are concatenated into a single token by the lexer. The
combined byte length is bounded by max_string_length (default
UINT16_MAX = 65535 bytes).

f) Number exponent in bare literals vs. quoted strings
Bare number literals only accept "e"/"E" as exponent markers
(exp_intro state).

g) Array row-size consistency
The "/"-separated dimension rows of a single array must all have
the same element count; the lexer checks each row as it closes, so
the violation is reported at the earliest offending byte. The
per-bracket-pair context save/restore mechanism keeps the /-row
check scoped to one array instance. In addition, array elements must
be homogeneous (spec 1.0, enforced above the grammar): sibling

```

- sub-arrays must match in length, so a bare array/matrix is rectangular.
Violation: `error_array_row_size_mismatch`
(e.g. `[1,2,3]/[4,5]`, `[[1,2]/[3,4,5]]`, or `[[1,2],[3,4,5]]`).
- h) Array and struct nesting limits
Array nesting (bracket depth): `max_array_nesting` (default 64 when the flag is 0; hard cap 255).
Struct nesting: `max_struct_nesting` (default 64 when the flag is 0; hard cap 255).
- i) Type-annotation / value compatibility (enforced by validator)
"utf8" requires a string value.
"uint", "sint", "float", "float_fix", "float_dec" accept a number or string value; a quoted string allows special float notation.
"utf8" and "bool" are parameterless: the lexer accepts the ":" and parameter bytes, but `bvn_parse_type_annotation` rejects any width, base, q, or unit parameter as `error_illegal_value_type` (e.g. "`<utf8:8>`" and "`<utf8:,m>`" are both errors).
"uint" rejects negative number literals (`error_value_out_of_range`).
Digit characters in the value must be valid for the declared base (`error_digit_not_in_base`).
The numeric value must fit in the declared bit-width (`error_value_out_of_range`).
For "float": accepted widths are 0, 16, or any multiple of 32 up to `BVN_FLOAT_MAX_PREC` (32768); accepted bases are 10 (or absent) and 16; any other base → `error_illegal_value_type`.
For "float_fix": accepted widths are 0, 16, 32, 64, 128, 256; the base parameter (`_N`) is forbidden (`error_illegal_value_type`); the Q parameter (`qN`) must satisfy $0 \leq Q < \text{effective_width}$ (`error_illegal_value_type` if violated); `Q` stored in `value_type_spec.t.base`; retrieved via `bvn_effective_q`.
For "float_dec": accepted widths are 0, 16, 32, 64, 128, 256; the base parameter (`_N`) is forbidden (`error_illegal_value_type`); the Q parameter is not valid for `float_dec`.
Neither "float_fix" nor "float_dec" can be synthesised by the validator's default type synthesis; both require explicit annotation.
- j) Identifier (key) non-empty
The grammar requires at least one id-start character after ".".
An empty key ("=") triggers `error_empty_identififer`.
- k) Struct closing brace without matching open
A "}" seen when `struct_nesting_level == 0` is `error_illegal_struct_close`.
- l) Comma in non-array context
"," triggers `ACT_new_array_value`, which checks `array_nesting_level > 0` and returns `error_unexpected_input_byte` if the condition is not met. A stray "," at the top level is therefore an error.
- m) Octet stream out-of-sync
Any byte other than 0x01 (chunk) or 0x00 (end) as the tag byte inside an octet stream causes `error_octet_stream_out_of_sync`.
- n) Error recovery (continue_on_error mode)
When `bvnr_read_flags_t.continue_on_error` is set, any parse error invokes the `on_error` callback and then enters the resync state machine (states: `resync`, `resync_string`, `resync_string_escape`, `resync_comment`). The resync machine skips bytes while tracking bracket and brace nesting depths, and attempts to re-synchronise at the next ";" at the current nesting depth. `recovery_count` (`bvnr_reader_get_recovery_count`) is incremented immediately when an error triggers entry into resync mode, not when resync completes at ";". EOF inside any resync state propagates the original error code rather than overwriting it with `error_got_incomplete_bvnr_stream`.

o) Inline unit suffix

An inline-unit follows a scalar number or string in non-array context only. A special-number keyword (nan, inf, ninf) takes NO inline unit – it is lexed as a symbol, and symbols accept no inline suffix; "inf m/s" is `error_unexpected_input_byte`. Supply a unit for a special value through the type annotation (`<float:64,m/s> inf`). At least one whitespace character MUST separate the value literal from the unit. The only accepted form is "value unit"; any other form (no separator or colon) is a hard error.

State-machine flow:

1. Whitespace from number-body states (`zero_intro`, `copy_number_byte`, `fraction_intro`, `copy_fraction_byte`, `copy_exp_byte`) fires `ACT_to_number_outro` → `number_outro`. Strings enter `string_outro_nosp` after the closing quote; whitespace transitions to `string_outro`. A comment's terminating newline also counts as whitespace (`comment_outro` promotes `string_outro_nosp` → `string_outro`).
2. From `number_outro` / `string_outro` (at least one ws consumed):
`alpha` / `"_"` / UTF-8-leader → `ACT_inline_unit_intro`
3. `ACT_inline_unit_intro` pushes the first byte and enters `inline_unit_body` directly.
4. `inline_unit_body` accumulates unit characters (including `":"` as a valid mid-unit byte) until whitespace (→ `inline_unit_outro`) or `;"` (→ `value_outro` via `finalize`).

Inside an array element (`in_array_element` flag set), `ACT_inline_unit_intro` immediately rejects with `error_unexpected_input_byte`.

The validator's `bn_val_receive` calls `bn_parse_unit` on the inline buffer and compares the result to `v->parsed_unit` using `memcmp` on the full `value_unit_t` structure (valid because both structs are zero-initialised before parsing and all padding is zeroed). When `has_annotation_unit` is true and the comparison fails, the validator returns `error_unit_mismatch`. When no annotation unit was present, the inline unit is silently adopted as the effective unit.

*)